A guide to the arithmetic certificate for $K = \mathbf{Q}(\sqrt{-61040707})$, $p = 5$

Generated by `tools/certificate_guide.gp`

What this document is

This guide restates the content of `certificate.gp` in the notation of Section 2.5 of the paper. Each of the 18 entries certifies one value $D_x(e_j)$ of a secondary norm operator: it stores the unramified cyclic quintic L_x/K, the normalized generator σ_x, the pair (a', J) representing the torsion class e_j, the auxiliary divisor I', the compactly represented element t, a residue prime for the sign check, and the expected norm class $[N_x(I')]$. The two defining equations are

$$(1 - \sigma_x)^2 I' + \operatorname{div}(t) + i_x(J) = 0, \qquad\qquad N_x(t) = a'.$$

Facts marked **Checked** are recomputed exactly by the generating script; the full verification is performed by the shared verifier `certificate/verify_certificate.c` as described in the README files of the `certificate` directory.

Notation and conventions

Ideals as matrices. An ideal of the ring of integers is stored by its *Hermite normal form* (HNF): an upper-triangular integer matrix whose columns are the coordinates, with respect to the fixed integral basis, of a $\mathbf{Z}$-basis of the ideal. For example, the ideal $J = \left(\begin{smallmatrix} 2713 & 751 \\ 0 & 1 \end{smallmatrix}\right)$ of Entry 1 below, over K with integral basis $(1, s)$, denotes the ideal $\mathbf{Z}\,2713 + \mathbf{Z}\,(751 + s)$. The determinant of the matrix is the norm of the ideal. Elements are likewise given by their coordinate vectors in the integral basis.

Which integral basis. The one PARI returns for the field in question – and that basis is LLL-reduced, so it is not canonical: another machine may reduce differently, and the same coordinate vector would then denote a different algebraic number. The certificate therefore records the basis it was written against, as its last field per entry and in the header for K. The verifier expresses the stored basis in its own, checks that the resulting matrix is integral with determinant ± 1 – so that both bases span the same ring of integers – and converts the coordinates before checking anything. Where the two bases agree, that matrix is the identity and nothing changes.

Characters. The header fixes three generators $\kappa_1, \kappa_2, \kappa_3$ of $\mathrm{Cl}(K)$ (as HNF ideals). A character x on $\mathrm{Cl}(K)/5$ is recorded by its value vector $(x(\bar\kappa_1), x(\bar\kappa_2), x(\bar\kappa_3))$; thus $(1, 0, 0)$ is the character dual to $\bar\kappa_1$. The column number j records which basis element e_j of $\mathrm{Cl}(K)[5]$ the stored pair (a', J) represents.

Automorphisms. σ_x is a field automorphism of L_x. Writing θ for the chosen root of the stored absolute polynomial, so that $L_x = \mathbf{Q}(\theta)$, the automorphism is determined by the single value $\sigma_x(\theta)$, and the certificate stores the coordinate vector of $\sigma_x(\theta)$ in the integral basis of L_x.

Base field data

The base field is $K = \mathbf{Q}(s)$ with $s^2 - s + 15260177 = 0$, of discriminant -61040707. Class group invariants $(20 \quad 10 \quad 5)$, class number 1000; the three stored generator ideals (HNF) are $\begin{pmatrix} 193 & 45 \\ 0 & 1 \end{pmatrix}$, $\begin{pmatrix} 761 & 748 \\ 0 & 1 \end{pmatrix}$, $\begin{pmatrix} 1213 & 296 \\ 0 & 1 \end{pmatrix}$, and the torsion units are generated by -1 of order 2.

Entry 1: character a, column e_1

Prescribed character vector $(1 \quad 0 \quad 0)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 - y^4 + (s + 596)\, y^3 + (-15s - 91547)\, y^2 + (-15s + 493029)\, y + (-594s + 4156434)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 - 2*y^9 + 1194*y^8 - 184302*y^7 + 16785141*y^6 - 559703590*y^5
+ 11937823312*y^4 - 96580412844*y^3 - 242586001017*y^2 + 43700661455
76*y + 22657814486532
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \quad 0 \quad 1 \quad 0 \quad 0 \quad 0 \quad 0 \quad 1 \quad 0 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{254164583}{146976861317244793} + \left(-\frac{73464}{146976861317244793}\right) s, \qquad J = \begin{pmatrix} 2713 & 751 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 2713$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$, equivalently $\mathrm{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $3223534230000 = 2^4 \cdot 3^2 \cdot 5^4 \cdot 7 \cdot 5116721$:

$$\begin{pmatrix}
2149022820 & 1785426780 & 1137977520 & 1142896165 & 1067502673 & 857198691 & 1784352178 & 1987745796 & 384106826 & 1259346964 \\
0 & 15 & 10 & 0 & 3 & 11 & 0 & 1 & 5 & 11 \\
0 & 0 & 20 & 15 & 11 & 4 & 1 & 0 & 12 & 0 \\
0 & 0 & 0 & 5 & 1 & 4 & 0 & 0 & 4 & 2 \\
0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1
\end{pmatrix}$$

The element t is stored in compact form as a product of 269 factors with signed exponents of absolute value up to 82539193764; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 11$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_1) = (0 \quad 1 \quad 0)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 2: character a, column e_2

Prescribed character vector $(1 \quad 0 \quad 0)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 - y^4 + (s + 596)\, y^3 + (-15s - 91547)\, y^2 + (-15s + 493029)\, y + (-594s + 4156434)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 - 2*y^9 + 1194*y^8 - 184302*y^7 + 16785141*y^6 - 559703590*y^5
+ 11937823312*y^4 - 96580412844*y^3 - 242586001017*y^2 + 43700661455
76*y + 22657814486532
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \quad 0 \quad 1 \quad 0 \quad 0 \quad 0 \quad 0 \quad 1 \quad 0 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{579396463}{533524229553878551} + \left(-\frac{113838}{533524229553878551}\right) s, \qquad J = \begin{pmatrix} 3511 & 1109 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 3511$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$, equivalently $\mathrm{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $1283346364716 = 2^2 \cdot 3^2 \cdot 13 \cdot 2742193087$:

$$\begin{pmatrix}
213891060786 & 35917654728 & 87186283752 & 3019265520 & 82720395321 & 164964566711 & 31410345170 & 16072540820 & 180702219631 & 127163484285 \\
0 & 3 & 2 & 2 & 0 & 2 & 2 & 0 & 0 & 0 \\
0 & 0 & 2 & 0 & 1 & 1 & 1 & 0 & 1 & 1 \\
0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1
\end{pmatrix}$$

The element t is stored in compact form as a product of 233 factors with signed exponents of absolute value up to 242462; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 11$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_2) = (0 \quad 4 \quad 3)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 3: character a, column e_3

Prescribed character vector $(1 \quad 0 \quad 0)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 - y^4 + (s + 596)\, y^3 + (-15s - 91547)\, y^2 + (-15s + 493029)\, y + (-594s + 4156434)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 - 2*y^9 + 1194*y^8 - 184302*y^7 + 16785141*y^6 - 559703590*y^5
+ 11937823312*y^4 - 96580412844*y^3 - 242586001017*y^2 + 43700661455
76*y + 22657814486532
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \quad 0 \quad 1 \quad 0 \quad 0 \quad 0 \quad 0 \quad 1 \quad 0 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{32972983}{2626056128537293} + (\frac{10043}{2626056128537293})\, s, \qquad J = \begin{pmatrix} 1213 & 296 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 1213$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$, equivalently $\mathrm{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $27418800636688 = 2^4 \cdot 19 \cdot 229 \cdot 7699 \cdot 51157$:

$$\begin{pmatrix}
3427350079586 & 2837092662522 & 2608895173368 & 2725223177634 & 273929779878 & 1666951123989 & 103912233536 & 721957175337 & 1303710000025 & 1625182099598 \\
0 & 2 & 0 & 0 & 0 & 1 & 0 & 1 & 0 & 0 \\
0 & 0 & 2 & 0 & 0 & 1 & 0 & 0 & 1 & 1 \\
0 & 0 & 0 & 2 & 0 & 0 & 0 & 1 & 0 & 0 \\
0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1
\end{pmatrix}$$

The element t is stored in compact form as a product of 274 factors with signed exponents of absolute value up to 84882662179; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 11$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_3) = (0 \quad 3 \quad 1)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 4: character b, column e_1

Prescribed character vector $(0 \quad 1 \quad 0)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 + 22y^3 + (22s - 11)\, y^2 + 2174656y + (-4480s + 2240)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 + 44*y^8 + 4349796*y^6 + 7481610411*y^4 + 1721042677376*y^2 + 3
06277851443200
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(1 \quad 2 \quad 0 \quad 1 \quad -1 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{254164583}{146976861317244793} + \left(-\frac{73464}{146976861317244793}\right) s, \qquad J = \begin{pmatrix} 2713 & 751 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 2713$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$, equivalently $\operatorname{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $37447973017268 = 2^2 \cdot 11^2 \cdot 193 \cdot 400890389$:

$$\begin{pmatrix}
1702180591694 & 258067167974 & 682605261550 & 694633268330 & 1349242601298 & 1074882142914 & 601175944021 & 671773640265 & 690264828372 & 735697201907 \\
0 & 22 & 0 & 12 & 2 & 11 & 21 & 1 & 9 & 18 \\
0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1
\end{pmatrix}$$

The element t is stored in compact form as a product of 203 factors with signed exponents of absolute value up to 2149542977247342950178242574753; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 19$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_1) = (4 \quad 0 \quad 4)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 5: character b, column e_2

Prescribed character vector $(0 \quad 1 \quad 0)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 + 22y^3 + (22s - 11)\, y^2 + 2174656y + (-4480s + 2240)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 + 44*y^8 + 4349796*y^6 + 7481610411*y^4 + 1721042677376*y^2 + 3
06277851443200
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(1 \quad 2 \quad 0 \quad 1 \quad -1 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{579396463}{533524229553878551} + \left(-\frac{113838}{533524229553878551}\right) s, \qquad J = \begin{pmatrix} 3511 & 1109 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 3511$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$, equivalently $\operatorname{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $13872356136639 = 3^2 \cdot 7^2 \cdot 11 \cdot 13 \cdot 2557 \cdot 86029$:

$$\begin{pmatrix}
660588387459 & 361590177312 & 4905090085 & 285548101967 & 261624281266 & 77192817684 & 281291431946 & 623907613571 & 460260605917 & 238076830007 \\
0 & 21 & 10 & 0 & 14 & 20 & 6 & 8 & 6 & 4 \\
0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1
\end{pmatrix}$$

The element t is stored in compact form as a product of 2 factors with signed exponents of absolute value up to 1; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 5$ with residue degree 2; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_2) = (4 \quad 0 \quad 1)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 6: character b, column e_3

Prescribed character vector $(0 \quad 1 \quad 0)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 + 22y^3 + (22s - 11)\, y^2 + 2174656y + (-4480s + 2240)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 + 44*y^8 + 4349796*y^6 + 7481610411*y^4 + 1721042677376*y^2 + 3
06277851443200
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(1 \quad 2 \quad 0 \quad 1 \quad -1 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{32972983}{2626056128537293} + \left(\frac{10043}{2626056128537293}\right) s, \qquad J = \begin{pmatrix} 1213 & 296 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 1213$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$, equivalently $\operatorname{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $18651079917568 = 2^{14} \cdot 7^2 \cdot 53 \cdot 438341$:

$$\begin{pmatrix} 5203984352 & 726604368 & 3963481616 & 1450420208 & 3784021470 & 5149115704 & 1033110481 & 4796696563 & 1621971298 & 2214930015 \\ 0 & 14 & 12 & 10 & 10 & 10 & 2 & 11 & 7 & 4 \\ 0 & 0 & 2 & 0 & 0 & 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 32 & 30 & 14 & 10 & 23 & 31 & 15 \\ 0 & 0 & 0 & 0 & 2 & 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 2 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 7 factors with signed exponents of absolute value up to 1; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 3$ with residue degree 2; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_3) = (4 \quad 0 \quad 3)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 7: character c, column e_1

Prescribed character vector $(0 \quad 0 \quad 1)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 - 3y^4 + (-s - 1149)\, y^3 + (-13s + 63831)\, y^2 + (908s - 547750)\, y + (10640s - 53110896)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 - 6*y^9 - 2290*y^8 + 134546*y^7 + 15103988*y^6 + 147104694*y^5
- 19483077347*y^4 - 632770881834*y^3 + 1880550308684*y^2 + 352989736
250912*y + 4547800708088576
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(1 \quad 0 \quad 1 \quad 1 \quad 0 \quad -1 \quad 0 \quad -1 \quad 0 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{254164583}{146976861317244793} + \left(-\frac{73464}{146976861317244793}\right) s, \qquad J = \begin{pmatrix} 2713 & 751 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 2713$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$, equivalently $\mathrm{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $36819507031524 = 2^2 \cdot 3^6 \cdot 7 \cdot 2713 \cdot 664879$:

$$\begin{pmatrix} 227280907602 & 151722996201 & 1762524141 & 99659990535 & 40823140242 & 41942221512 & 88198242124 & 141706878240 & 108060483270 & 14985868604 \\ 0 & 9 & 3 & 5 & 3 & 5 & 8 & 0 & 8 & 4 \\ 0 & 0 & 6 & 1 & 3 & 5 & 5 & 2 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 3 & 1 & 1 & 2 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 361 factors with signed exponents of absolute value up to 45156465607541555777761446960; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 11$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_1) = (3 \quad 0 \quad 0)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 8: character c, column e_2

Prescribed character vector $(0 \quad 0 \quad 1)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 - 3y^4 + (-s - 1149)\,y^3 + (-13s + 63831)\,y^2 + (908s - 547750)\,y + (10640s - 53110896)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 - 6*y^9 - 2290*y^8 + 134546*y^7 + 15103988*y^6 + 147104694*y^5
- 19483077347*y^4 - 632770881834*y^3 + 1880550308684*y^2 + 352989736
250912*y + 4547800708088576
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(1 \quad 0 \quad 1 \quad 1 \quad 0 \quad -1 \quad 0 \quad -1 \quad 0 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{579396463}{533524229553878551} + (-\frac{113838}{533524229553878551})\,s, \qquad J = \begin{pmatrix} 3511 & 1109 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 3511$. **Checked:** $(a')\,J^5 = \mathcal{O}_K$, equivalently $\mathrm{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $7963940593728 = 2^6 \cdot 3^2 \cdot 7^4 \cdot 277 \cdot 20789$:

$$\begin{pmatrix} 483718452 & 227867976 & 132130880 & 211282414 & 410684344 & 49693198 & 120113279 & 422541736 & 120815308 & 141680722 \\ 0 & 84 & 70 & 42 & 21 & 0 & 28 & 70 & 19 & 32 \\ 0 & 0 & 14 & 0 & 8 & 8 & 13 & 6 & 2 & 3 \\ 0 & 0 & 0 & 14 & 7 & 8 & 1 & 0 & 1 & 12 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 359 factors with signed exponents of absolute value up to 38013778069801827058833375 36; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 11$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_2) = (1 \quad 0 \quad 0)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 9: character c, column e_3

Prescribed character vector $(0 \quad 0 \quad 1)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 - 3y^4 + (-s - 1149)\, y^3 + (-13s + 63831)\, y^2 + (908s - 547750)\, y + (10640s - 53110896)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 - 6*y^9 - 2290*y^8 + 134546*y^7 + 15103988*y^6 + 147104694*y^5
- 19483077347*y^4 - 632770881834*y^3 + 1880550308684*y^2 + 352989736
250912*y + 4547800708088576
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(1 \quad 0 \quad 1 \quad 1 \quad 0 \quad -1 \quad 0 \quad -1 \quad 0 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{32972983}{2626056128537293} + (\frac{10043}{2626056128537293})\, s, \qquad J = \begin{pmatrix} 1213 & 296 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 1213$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$, equivalently $\mathrm{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $25661276390144 = 2^8 \cdot 107 \cdot 7309 \cdot 128173$:

$$\begin{pmatrix} 400957443596 & 13388486968 & 231719016696 & 331499651696 & 55411858338 & 308723671139 & 102041695281 & 1934600540 & 346085036866 & 292247556734 \\ 0 & 4 & 0 & 0 & 0 & 0 & 1 & 1 & 0 & 2 \\ 0 & 0 & 4 & 0 & 1 & 0 & 1 & 1 & 2 & 3 \\ 0 & 0 & 0 & 4 & 1 & 0 & 3 & 3 & 3 & 1 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 300 factors with signed exponents of absolute value up to 38367884847714121030263148024; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 11$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_3) = (2 \quad 2 \quad 0)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 10: character $a + b$, column e_1

Prescribed character vector $(1 \quad 1 \quad 0)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 - y^4 + (-s - 849)\, y^3 + (15s + 52354)\, y^2 + (-159s - 1104054)\, y + (-2150s + 5958260)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 - 2*y^9 - 1698*y^8 + 106422*y^7 + 13668837*y^6 - 532644854*y^5
+ 12892011104*y^4 - 132921712908*y^3 + 1244476198619*y^2 - 272167220
4420*y + 106028220151100
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \quad 0 \quad -1 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0 \quad 1 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{254164583}{146976861317244793} + \left(-\frac{73464}{146976861317244793}\right) s, \qquad J = \begin{pmatrix} 2713 & 751 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 2713$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$, equivalently $\mathrm{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $42055285409564 = 2^2 \cdot 7 \cdot 1501974478913$:

$$\begin{pmatrix}
21027642704782 & 6687803274550 & 2206557951661 & 15759836842411 & 9116092259351 & 2531545033149 & 4391281621681 & 16923864274922 & 7455345516788 & 12073422506684 \\
0 & 2 & 0 & 0 & 0 & 1 & 0 & 1 & 1 & 1 \\
0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1
\end{pmatrix}$$

The element t is stored in compact form as a product of 320 factors with signed exponents of absolute value up to 76540389117; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 11$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_1) = (3 \quad 2 \quad 2)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 11: character $a+b$, column e_2

Prescribed character vector $(1 \quad 1 \quad 0)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 - y^4 + (-s - 849)\, y^3 + (15s + 52354)\, y^2 + (-159s - 1104054)\, y + (-2150s + 5958260)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 - 2*y^9 - 1698*y^8 + 106422*y^7 + 13668837*y^6 - 532644854*y^5
+ 12892011104*y^4 - 132921712908*y^3 + 1244476198619*y^2 - 272167220
4420*y + 106028220151100
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \quad 0 \quad -1 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0 \quad 1 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{579396463}{533524229553878551} + \left(-\frac{113838}{533524229553878551}\right) s, \qquad J = \begin{pmatrix} 3511 & 1109 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 3511$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$, equivalently $\mathrm{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $61701674840064 = 2^{12} \cdot 3^2 \cdot 7 \cdot 31^2 \cdot 248813$:

$$\begin{pmatrix}
1295818104 & 366026052 & 1161697736 & 793935487 & 479271616 & 362044164 & 694620372 & 351302106 & 959487044 & 1245163039 \\
0 & 372 & 8 & 136 & 250 & 289 & 312 & 124 & 41 & 117 \\
0 & 0 & 8 & 4 & 2 & 0 & 4 & 0 & 0 & 5 \\
0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 2 & 1 & 0 & 0 & 0 & 1 \\
0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 4 & 0 & 0 & 3 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 2 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1
\end{pmatrix}$$

The element t is stored in compact form as a product of 320 factors with signed exponents of absolute value up to 7068151712; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 11$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_2) = (0 \quad 0 \quad 4)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 12: character $a+b$, column e_3

Prescribed character vector $(1 \quad 1 \quad 0)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 - y^4 + (-s - 849)\, y^3 + (15s + 52354)\, y^2 + (-159s - 1104054)\, y + (-2150s + 5958260)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 - 2*y^9 - 1698*y^8 + 106422*y^7 + 13668837*y^6 - 532644854*y^5
+ 12892011104*y^4 - 132921712908*y^3 + 1244476198619*y^2 - 272167220
4420*y + 106028220151100
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \quad 0 \quad -1 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0 \quad 1 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{32972983}{2626056128537293} + (\frac{10043}{2626056128537293})\, s, \qquad J = \begin{pmatrix} 1213 & 296 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 1213$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$, equivalently $\operatorname{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $11640919262427 = 3^2 \cdot 7 \cdot 13 \cdot 61 \cdot 139 \cdot 953 \cdot 1759$:

$$\begin{pmatrix}
3880306420809 & 2770947584832 & 257541559789 & 646517670096 & 2228094089263 & 1694106045929 & 124414833767 & 539641311615 & 2306317086218 & 3135086743723 \\
0 & 3 & 1 & 0 & 1 & 0 & 2 & 2 & 2 & 0 \\
0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1
\end{pmatrix}$$

The element t is stored in compact form as a product of 319 factors with signed exponents of absolute value up to 149665316424; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 11$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_3) = (1 \quad 4 \quad 1)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 13: character $a + c$, column e_1

Prescribed character vector $(1 \quad 0 \quad 1)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 - y^4 + 532y^3 + (-16s + 80745)\, y^2 + (298s + 716171)\, y + (-15842s + 71652976)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 - 2*y^9 + 1065*y^8 + 160410*y^7 + 1554190*y^6 + 227761638*y^5 +
11043942787*y^4 + 46376348712*y^3 + 21173147731929*y^2 - 4144245356
8606*y + 8962844655357412
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \quad -1 \quad 0 \quad -1 \quad 0 \quad 0 \quad 0 \quad 1 \quad 0 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{254164583}{146976861317244793} + \left(-\frac{73464}{146976861317244793}\right) s, \qquad J = \begin{pmatrix} 2713 & 751 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 2713$. **Checked:** $(a')\,J^5 = \mathcal{O}_K$, equivalently $\mathrm{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $1413524858297 = 29 \cdot 48742236493$:

$$\begin{pmatrix} 1413524858297 & 694132558939 & 224372580134 & 570836977290 & 920159279216 & 1151675387126 & 78209000894 & 344919351288 & 211108289585 & 1005992030845 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 301 factors with signed exponents of absolute value up to 1440721437012; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 11$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_1) = (3 \quad 1 \quad 2)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 14: character $a + c$, column e_2

Prescribed character vector $(1 \quad 0 \quad 1)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 - y^4 + 532y^3 + (-16s + 80745)\,y^2 + (298s + 716171)\,y + (-15842s + 71652976)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 - 2*y^9 + 1065*y^8 + 160410*y^7 + 1554190*y^6 + 227761638*y^5 +
11043942787*y^4 + 46376348712*y^3 + 21173147731929*y^2 - 4144245356
8606*y + 8962844655357412
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \quad -1 \quad 0 \quad -1 \quad 0 \quad 0 \quad 0 \quad 1 \quad 0 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{579396463}{533524229553878551} + (-\frac{113838}{533524229553878551})\, s, \qquad J = \begin{pmatrix} 3511 & 1109 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 3511$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$, equivalently $\mathrm{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $32646359966976 = 2^8 \cdot 3^2 \cdot 7^5 \cdot 843067$:

$$\begin{pmatrix}
991446792 & 232200906 & 648221154 & 384666779 & 652330094 & 133447866 & 763883919 & 614873529 & 920565661 & 928259404 \\
0 & 294 & 238 & 130 & 260 & 42 & 68 & 247 & 282 & 17 \\
0 & 0 & 56 & 31 & 20 & 12 & 40 & 46 & 10 & 50 \\
0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 2 & 0 & 1 & 0 & 1 \\
0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1
\end{pmatrix}$$

The element t is stored in compact form as a product of 241 factors with signed exponents of absolute value up to 166439366787; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 11$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_2) = (3 \quad 3 \quad 2)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 15: character $a + c$, column e_3

Prescribed character vector $(1 \quad 0 \quad 1)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 - y^4 + 532y^3 + (-16s + 80745)\, y^2 + (298s + 716171)\, y + (-15842s + 71652976)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 - 2*y^9 + 1065*y^8 + 160410*y^7 + 1554190*y^6 + 227761638*y^5 +
11043942787*y^4 + 46376348712*y^3 + 21173147731929*y^2 - 4144245356
8606*y + 8962844655357412
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \quad -1 \quad 0 \quad -1 \quad 0 \quad 0 \quad 0 \quad 1 \quad 0 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{32972983}{2626056128537293} + (\frac{10043}{2626056128537293})\, s, \qquad J = \begin{pmatrix} 1213 & 296 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 1213$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$, equivalently $\mathrm{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $19500028948944 = 2^4 \cdot 3^2 \cdot 135416867701$:

$$\begin{pmatrix} 812501206206 & 386452438650 & 442758507896 & 704577392328 & 3164248961 & 68346854312 & 340990993512 & 606640849321 & 548026715662 & 419588420064 \\ 0 & 6 & 2 & 4 & 3 & 2 & 1 & 3 & 4 & 0 \\ 0 & 0 & 2 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 2 & 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 231 factors with signed exponents of absolute value up to 271011566; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 11$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_3) = (1 \quad 2 \quad 4)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 16: character $b + c$, column e_1

Prescribed character vector $(0 \quad 1 \quad 1)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 - 2y^4 + (-s - 1832)\, y^3 + (22s - 19966)\, y^2 + (-93s + 666465)\, y + (-360s - 1976130)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 - 4*y^9 - 3661*y^8 - 32580*y^7 + 20030890*y^6 - 604930996*y^5 +
8188001885*y^4 - 70810902186*y^3 + 413252218233*y^2 - 1612277656290
*y + 5883520122900
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(1 \quad 1 \quad 0 \quad -1 \quad 1 \quad 0 \quad 0 \quad 0 \quad -1 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{254164583}{146976861317244793} + \left(-\frac{73464}{146976861317244793}\right) s, \qquad J = \begin{pmatrix} 2713 & 751 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 2713$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$, equivalently $\mathrm{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $28278521334297 = 3^8 \cdot 13 \cdot 193 \cdot 1717853$:

$$\begin{pmatrix} 38790838593 & 6164566551 & 24904683843 & 21111862527 & 31847022519 & 32507928810 & 36947507045 & 23369648839 & 10067395646 & 26213362722 \\ 0 & 9 & 0 & 6 & 6 & 3 & 4 & 0 & 8 & 3 \\ 0 & 0 & 3 & 0 & 0 & 0 & 2 & 1 & 0 & 2 \\ 0 & 0 & 0 & 3 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 3 & 0 & 0 & 2 & 2 & 1 \\ 0 & 0 & 0 & 0 & 0 & 3 & 1 & 2 & 2 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 382 factors with signed exponents of absolute value up to 6194921944653; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 11$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_1) = (4 \quad 1 \quad 4)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 17: character $b + c$, column e_2

Prescribed character vector $(0 \quad 1 \quad 1)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 - 2y^4 + (-s - 1832)\,y^3 + (22s - 19966)\,y^2 + (-93s + 666465)\,y + (-360s - 1976130)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 - 4*y^9 - 3661*y^8 - 32580*y^7 + 20030890*y^6 - 604930996*y^5 +
8188001885*y^4 - 70810902186*y^3 + 413252218233*y^2 - 1612277656290
*y + 5883520122900
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(1 \quad 1 \quad 0 \quad -1 \quad 1 \quad 0 \quad 0 \quad 0 \quad -1 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{579396463}{533524229553878551} + \left(-\frac{113838}{533524229553878551}\right) s, \qquad J = \begin{pmatrix} 3511 & 1109 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 3511$. **Checked:** $(a')\,J^5 = \mathcal{O}_K$, equivalently $\mathrm{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $5239772081804 = 2^2 \cdot 13 \cdot 5039 \cdot 19996993$:

$$\begin{pmatrix} 2619886040902 & 1904951301564 & 1376921369383 & 1881009665917 & 827239693275 & 970391940115 & 1925408882982 & 962382564726 & 720046357498 & 551001009085 \\ 0 & 2 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 381 factors with signed exponents of absolute value up to 5997408430356; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 11$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_2) = \begin{pmatrix}1 & 4 & 1\end{pmatrix}$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 18: character $b + c$, column e_3

Prescribed character vector $\begin{pmatrix}0 & 1 & 1\end{pmatrix}$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 - 2y^4 + (-s - 1832)\,y^3 + (22s - 19966)\,y^2 + (-93s + 666465)\,y + (-360s - 1976130)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 - 4*y^9 - 3661*y^8 - 32580*y^7 + 20030890*y^6 - 604930996*y^5 +
8188001885*y^4 - 70810902186*y^3 + 413252218233*y^2 - 1612277656290
*y + 5883520122900
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $\begin{pmatrix}1 & 1 & 0 & -1 & 1 & 0 & 0 & 0 & -1 & 0\end{pmatrix}$.

The pair (a', J) representing the torsion class

$$a' = -\frac{32972983}{2626056128537293} + \left(\frac{10043}{2626056128537293}\right) s, \qquad J = \begin{pmatrix}1213 & 296\\ 0 & 1\end{pmatrix}.$$

$N(J) = 1213$. **Checked:** $(a')\,J^5 = \mathcal{O}_K$, equivalently $\mathrm{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $31211851870487 = 31211851870487$:

$$\begin{pmatrix}
31211851870487 & 7586404893306 & 26914376810131 & 18061524010582 & 22716375521994 & 8315799643443 & 18488745765277 & 24329618889961 & 18022785923427 & 17885460220108\\
0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0\\
0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0\\
0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0\\
0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0\\
0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0\\
0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0\\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0\\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0\\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1
\end{pmatrix}$$

The element t is stored in compact form as a product of 376 factors with signed exponents of absolute value up to 1884749538298; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 11$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_3) = \begin{pmatrix}2 & 4 & 1\end{pmatrix}$ in the coordinates of $\mathrm{Cl}(K)/5$.